Experiment No. 1

Aim:

To install Keil µVision IDE and Flash Magic software for developing, compiling, and programming embedded C applications on microcontrollers.

Objectives:

- To install Keil µVision Integrated Development Environment (IDE).
- To install Flash Magic programming software.
- To configure the development environment for microcontroller programming.
- To verify the successful installation of both software tools.

Apparatus / Software Required :

- Computer with Windows 10/11 Operating System
- Internet Connection
- Keil µVision IDE
- Flash Magic Software
- USB Cable / Serial Cable (for programming)
- Microcontroller Development Board (Optional)

Theory:

Keil µVision

Keil µVision is a software tool used to develop programs for small devices like microcontrollers. It allows users to write, compile, test, and debug code written in C or assembly language for different types of microcontrollers, especially those based on the 8051 and ARM architectures.This tool brings together important features like a code writing window, project management, a compiler, and debugging tools into one easy-to-use program, making the process of creating software for these devices more efficient.

Flash Magic

Flash Magic is a tool used to transfer compiled code, usually in a file format called HEX, into the memory of a microcontroller.

It connects to the device through a serial or USB connection and has functions for erasing existing data, writing new program code, and checking that the data was written correctly.This ensures that the device works properly with the new software.

Procedure:

Part A: Installing Keil µVision

Step 1: Downloading the Software

1. Open your preferred web browser.
2. Go to the official Keil website.
3. Find and download the correct version of Keil µVision, like C51 or MDK-ARM, depending on your needs.

Step 2: Installing Keil µVision

1. Find the downloaded installer file and double-click to open it.
2. Click the "Next" button to continue.
3. Read the license agreement and click "I Agree" to accept it.
4. Choose where you want to install the software or use the default location.
5. Click "Install" to start the setup.
6. Wait until the installation is done.
7. Click "Finish" to close the installer.

Step 3: Launching Keil µVision

1. Start Keil µVision from the Start menu or the desktop icon.
2. Make sure the program opens properly and shows the interface without any problems.

Part B: Installing Flash Magic

Step 1: Downloading Flash Magic

1. Use a web browser to visit the official Flash Magic website.
2. Download the latest version of the Flash Magic software.

Step 2: Installing Flash Magic

1. Run the downloaded setup file.
2. Click "Next" to go through the installation steps.
3. Read the license agreement and click "I Agree" to accept it.
4. Choose a folder for the software installation.
5. Click "Install" to start the installation process.
6. Wait until it is finished.
7. Click "Finish" to exit the installer.

Step 3: Launching Flash Magic

1. Open Flash Magic from where it was installed.
2. Check that the program starts correctly and is ready to use.

Configuring Flash Magic for Programming

1. Connect your microcontroller development board to the computer with the right cable.
2. In the Flash Magic window, set up these options:
    - The type of microcontroller being used
    - The COM port it is connected to
    - The communication speed (baud rate)
    - The oscillator frequency of the microcontroller
3. Use the file browser to find and choose the HEX file you want to upload.
4. Click the "Start" button to begin the programming process.

Conclusion :

Keil uVision IDE was successfully installed and configured for use. The development is ready for editing,compiling and running embedded C applications.